Murphy NH3(g) Data

$A = 27.31$

$B = 0.02383$

$C = 1.707 \cdot 10^{-5}$

$D = -1.185 \cdot 10^{-8}$

$Cp = A + B\,T + C\,T^2 + D\,T^3$

$$Cp = -1.185\frac{T^3}{10^8} + 1.707\frac{T^2}{10^5} + 0.02383\,T + 27.31$$ *Substitute*

$$\Delta H = \frac{1}{1000}\left(\int_{298}^{473} Cp\ dT\right)$$

$$\Delta H = \frac{1}{1000}\left(-0.29625\frac{473^4}{10^8} + 0.569\frac{473^3}{10^5} + 0.29625\frac{298^4}{10^8} - 0.569\frac{298^3}{10^5} + 0.011915 \cdot 473^2 - 0.011915 \cdot 298^2 + 4779.3\right)$$ *Substitute*

$\Delta H = 6.7135$ *Calculate*

Murphy O2(g) Data

$A = 28.11$

$B = -3.68 \cdot 10^{-6}$

$C = 1.746 \cdot 10^{-5}$

$D = -1.065 \cdot 10^{-8}$

$Cp = A + B\,T + C\,T^2 + D\,T^3$

$$Cp = -3.68\frac{T}{10^6} - 1.065\frac{T^3}{10^8} + 1.746\frac{T^2}{10^5} + 28.11$$ *Substitute*

$$\Delta H = \frac{1}{1000}\left(\int_{298}^{473} Cp\ dT\right)$$

$$\Delta H = \frac{1}{1000}\left(-0.26625\frac{473^4}{10^8} + 0.582\frac{473^3}{10^5} - 1.84\frac{473^2}{10^6} + 0.26625\frac{298^4}{10^8} - 0.582\frac{298^3}{10^5} + 1.84\frac{298^2}{10^6} + 4919.2\right)$$ *Substitute*

$\Delta H = 5.2686$ *Calculate*

Murphy O2(g) Data

$A = 28.11$

$B = -3.68 \cdot 10^{-6}$

$C = 1.746 \cdot 10^{-5}$

$D = -1.065 \cdot 10^{-8}$

$\mathrm{Cp} = A + B\,T + C\,T^2 + D\,T^3$

$\mathrm{Cp} = -3.68\frac{T}{10^6} - 1.065\frac{T^3}{10^8} + 1.746\frac{T^2}{10^5} + 28.11$ *Substitute*

$\Delta\mathrm{H} = \frac{1}{1000}\left(\int_{298}^{T} \mathrm{Cp}\ dT\right)$

$\Delta\mathrm{H} = \frac{1}{1000}\left(-0.26625\frac{T^4}{10^8} + 0.582\frac{T^3}{10^5} - 1.84\frac{T^2}{10^6} + 28.11\,T + 0.26625\frac{298^4}{10^8} - 0.582\frac{298^3}{10^5} + 1.84\frac{298^2}{10^6} - 8376.8\right)$

$\Delta\mathrm{H} = 0.001\left(-2.6625\times10^{-9}\,T^4 + 5.82\times10^{-6}\,T^3 - 1.84\times10^{-6}\,T^2 + 28.11\,T - 8509.6\right)$ *Calculate*

$\Delta\mathrm{H} = -2.6625\times10^{-12}\,T^4 + 5.82\times10^{-9}\,T^3 - 1.84\times10^{-9}\,T^2 + 0.02811\,T - 8.5096$ *Expand*

Murphy NO(g) Data

$A = 29.35$

$B = -9.378 \cdot 10^{-4}$

$C = 9.747 \cdot 10^{-6}$

$D = -4.187 \cdot 10^{-9}$

$\mathrm{Cp} = A + B\,T + C\,T^2 + D\,T^3$

$\mathrm{Cp} = -4.187\frac{T^3}{10^9} + 9.747\frac{T^2}{10^6} - 0.0009378\,T + 29.35$ *Substitute*

$\Delta\mathrm{H} = \frac{1}{1000}\left(\int_{298}^{T} \mathrm{Cp}\ dT\right)$

$\Delta\mathrm{H} = \frac{1}{1000}\left(-1.0468\frac{T^4}{10^9} + 3.249\frac{T^3}{10^6} - 0.0004689\,T^2 + 29.35\,T + 1.0468\frac{298^4}{10^9} - 3.249\frac{298^3}{10^6} + 0.0004689 \cdot 298^2 - 8746.3\right)$ *Substit*

$\Delta\mathrm{H} = 0.001\left(-1.0468\times10^{-9}\,T^4 + 3.249\times10^{-6}\,T^3 - 0.0004689\,T^2 + 29.35\,T - 8782.4\right)$ *Calculate*

$\Delta\mathrm{H} = -1.0468\times10^{-12}\,T^4 + 3.249\times10^{-9}\,T^3 - 4.689\times10^{-7}\,T^2 + 0.02935\,T - 8.7824$ *Expand*

Murphy H2O(g) Data

$A = 32.24$

$B = 0.001924$

$C = 1.005 \cdot 10^{-5}$

$D = -3.59 \cdot 10^{-9}$

$\mathrm{Cp} = A + B\,T + C\,T^2 + D\,T^3$

$\mathrm{Cp} = -3.59\frac{T^3}{10^9} + 1.005\frac{T^2}{10^5} + 0.001924\,T + 32.24$ *Substitute*

$\Delta\mathrm{H} = \frac{1}{1000}\left(\int_{298}^{T} \mathrm{Cp}\; dT\right)$

$\Delta\mathrm{H} = \frac{1}{1000}\left(-0.8975\frac{T^4}{10^9} + 0.335\frac{T^3}{10^5} + 0.000962\,T^2 + 32.24\,T + 0.8975\frac{298^4}{10^9} - 0.335\frac{298^3}{10^5} - 0.000962 \cdot 298^2 - 9607.5\right)$

$\Delta\mathrm{H} = 0.001\left(-8.975\times10^{-10}\,T^4 + 3.35\times10^{-6}\,T^3 + 0.000962\,T^2 + 32.24\,T - 9774.5\right)$ *Calculate*

$\Delta\mathrm{H} = -8.975\times10^{-13}\,T^4 + 3.35\times10^{-9}\,T^3 + 9.62\times10^{-7}\,T^2 + 0.03224\,T - 9.7745$ *Expand*

Energy Balance

$\Delta\mathrm{Ha} = 6.7135$

$\Delta\mathrm{Hb} = 5.2686$

$\Delta\mathrm{Hc} = -2.6625\times10^{-12}\,T^4 + 5.82\times10^{-9}\,T^3 - 1.84\times10^{-9}\,T^2 + 0.02811\,T - 8.5096$

$\Delta\mathrm{Hd} = -1.0468\times10^{-12}\,T^4 + 3.249\times10^{-9}\,T^3 - 4.689\times10^{-7}\,T^2 + 0.02935\,T - 8.7824$

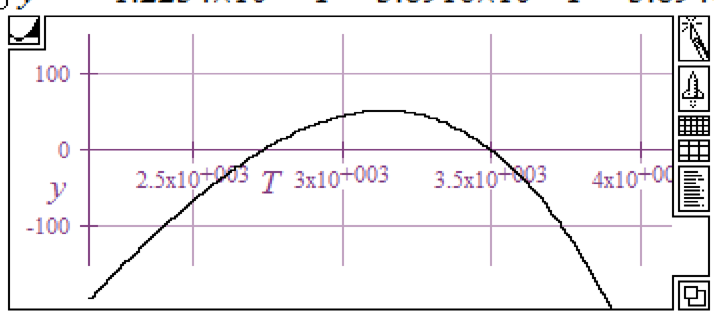
$\Delta\mathrm{He} = -8.975\times10^{-13}\,T^4 + 3.35\times10^{-9}\,T^3 + 9.62\times10^{-7}\,T^2 + 0.03224\,T - 9.7745$

$0 = 1(-905.9) + 1\,\Delta\mathrm{Hc} + 4\,\Delta\mathrm{Hd} + 6\,\Delta\mathrm{He} - 4\,\Delta\mathrm{Ha} - 6\,\Delta\mathrm{Hb}$

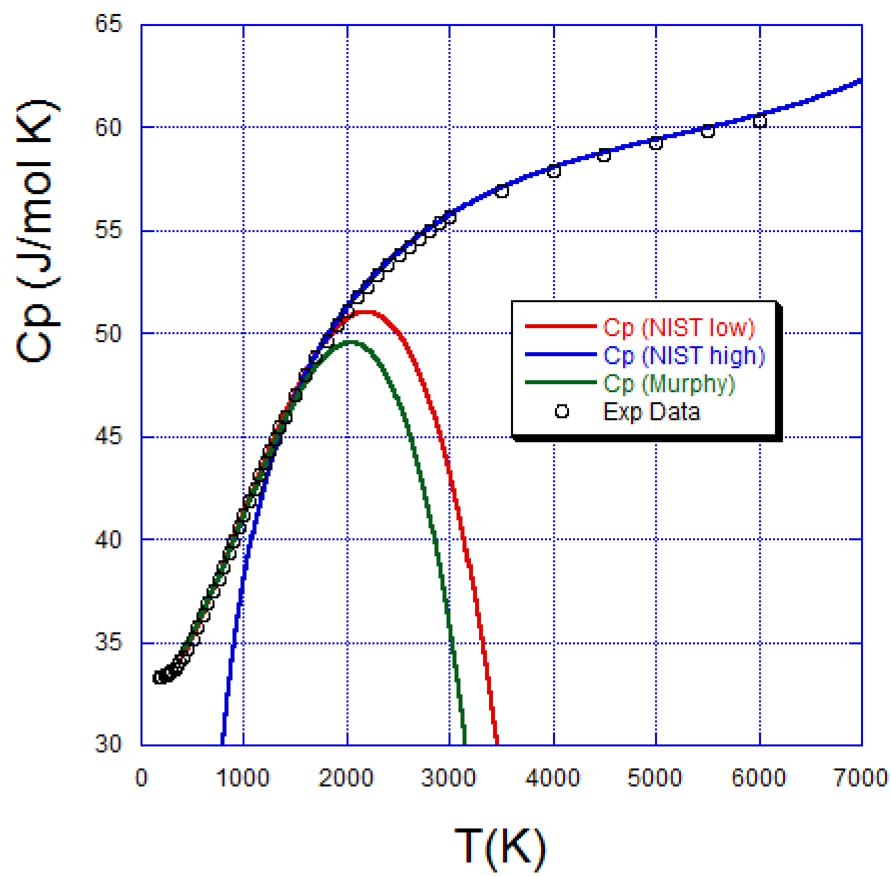
$0 = 6\left(-8.975\times10^{-13}\,T^4 + 3.35\times10^{-9}\,T^3 + 9.62\times10^{-7}\,T^2 + 0.03224\,T - 9.7745\right) + 4(-1.$

$0 = -1.2234\times10^{-11}\,T^4 + 3.8916\times10^{-8}\,T^3 + 3.8946\times10^{-6}\,T^2 + 0.33895\,T - 1066.7$

$y = -1.2234\times10^{-11}\,T^4 + 3.8916\times10^{-8}\,T^3 + 3.8946\times10^{-6}\,T^2 + 0.33895\,T - 1066.7$



Heat Capcity of Water Vapor
NIST vs. Murphy



NIST Oxygen (O2) Data

Gas Phase Heat Capacity (Shomate Equation)

$C_p° = A + B*t + C*t^2 + D*t^3 + E/t^2$
$H° − H°_{298.15} = A*t + B*t^2/2 + C*t^3/3 + D*t^4/4 − E/t + F − H$
$S° = A*\ln(t) + B*t + C*t^2/2 + D*t^3/3 − E/(2*t^2) + G$
C_p = heat capacity (J/mol*K)
H° = standard enthalpy (kJ/mol)
S° = standard entropy (J/mol*K)
t = temperature (K) / 1000.

View plot Requires a JavaScript / HTML 5 canvas capable browser.

View table.

Temperature (K)	100. - 700.	700. - 2000.	2000. - 6000.
A	31.32234	30.03235	20.91111
B	-20.23531	8.772972	10.72071
C	57.86644	-3.988133	-2.020498
D	-36.50624	0.788313	0.146449
E	-0.007374	-0.741599	9.245722
F	-8.903471	-11.32468	5.337651
G	246.7945	236.1663	237.6185
H	0.0	0.0	0.0
Reference	Chase, 1998	Chase, 1998	Chase, 1998
Comment	Data last reviewed in March, 1977; New parameter fit January 2009	Data last reviewed in March, 1977; New parameter fit January 2009	Data last reviewed in March, 1977; New parameter fit January 2009

NIST Nitric Oxide (NO) Data

Gas Phase Heat Capacity (Shomate Equation)

$C_p° = A + B*t + C*t^2 + D*t^3 + E/t^2$
$H° − H°_{298.15} = A*t + B*t^2/2 + C*t^3/3 + D*t^4/4 − E/t + F − H$
$S° = A*\ln(t) + B*t + C*t^2/2 + D*t^3/3 − E/(2*t^2) + G$

C_p = heat capacity (J/mol*K)
H° = standard enthalpy (kJ/mol)
S° = standard entropy (J/mol*K)
t = temperature (K) / 1000.

View plot Requires a JavaScript / HTML 5 canvas capable browser.

View table.

Temperature (K)	298. - 1200.	1200. - 6000.
A	23.83491	35.99169
B	12.58878	0.957170
C	-1.139011	-0.148032
D	-1.497459	0.009974
E	0.214194	-3.004088
F	83.35783	73.10787
G	237.1219	246.1619
H	90.29114	90.29114
Reference	Chase, 1998	Chase, 1998
Comment	Data last reviewed in June, 1963	Data last reviewed in June, 1963

NIST Water Vapor Data

Gas Phase Heat Capacity (Shomate Equation)

$C_p° = A + B*t + C*t^2 + D*t^3 + E/t^2$

$H° − H°_{298.15} = A*t + B*t^2/2 + C*t^3/3 + D*t^4/4 − E/t + F − H$

$S° = A*\ln(t) + B*t + C*t^2/2 + D*t^3/3 − E/(2*t^2) + G$

C_p = heat capacity (J/mol*K)

H° = standard enthalpy (kJ/mol)

S° = standard entropy (J/mol*K)

t = temperature (K) / 1000.

View plot Requires a JavaScript / HTML 5 canvas capable browser.

View table.

Temperature (K)	500. - 1700.	1700. - 6000.
A	30.09200	41.96426
B	6.832514	8.622053
C	6.793435	-1.499780
D	-2.534480	0.098119
E	0.082139	-11.15764
F	-250.8810	-272.1797
G	223.3967	219.7809
H	-241.8264	-241.8264
Reference	Chase, 1998	Chase, 1998
Comment	Data last reviewed in March, 1979	Data last reviewed in March, 1979

$$Ha = 6.7048$$

$$Hb = 5.253$$

$$Hc = -9.2457\frac{1}{t} + 0.036612\,t^4 - 0.6735\,t^3 + 5.3604\,t^2 + 20.911\,t + 5.3377$$

$$Hd = 3.0041\frac{1}{t} + 0.0024935\,t^4 - 0.049343\,t^3 + 0.47858\,t^2 + 35.992\,t - 17.183$$

$$He = 11.158\frac{1}{t} + 0.02453\,t^4 - 0.49993\,t^3 + 4.311\,t^2 + 41.964\,t - 30.354$$

$$0 = -905.9 + (1\,Hc + 4\,Hd + 6\,He) - (4\,Ha + 6\,Hb)$$

$$0 = 4\left(3.0041\frac{1}{t} + 0.0024935\,t^4 - 0.049343\,t^3 + 0.47858\,t^2 + 35.992\,t - 17.183\right)$$
$$+6\left(11.158\frac{1}{t} + 0.02453\,t^4 - 0.49993\,t^3 + 4.311\,t^2 + 41.964\,t - 30.354\right) - 9.2457\frac{1}{t} + 0.036612\,t^4 - 0.6735\,t^3 + 5.3604\,t^2 + 20.911\,t - 958.9$$

$$0 = 69.716\frac{1}{t} + 0.19376\,t^4 - 3.8704\,t^3 + 33.141\,t^2 + 416.66\,t - 1209.8$$ *Expand*

$$y = 69.716\frac{1}{t} + 0.19376\,t^4 - 3.8704\,t^3 + 33.141\,t^2 + 416.66\,t - 1209.8$$